

look less like a simple program and more like a textbook, with step-by-step explanations and visualisations of what the code does.

One of the major benefits of Google Colab is that it supports development in both Python2 and Python3. Currently, it does not support the R programming language. Google Colab supports the installation and import of all the major machine learning libraries such as Numpy, TensorFlow, PyTorch, Scikitlearn, Matplotlib, Seaborn, Pandas, etc.

You can save the Ipython notebooks that you create straight to your Google Drive, thus making sure your data is always available on the cloud. Another major benefit is the ease with which you can share your notebooks with anyone online. You can collaborate with your peers remotely on Google Colab by setting the appropriate permissions for others editing your notebook.

Setting up Google Colab

Creating a notebook on Google Colab is super easy. All you need is a Google account. Just follow the steps listed below.

- 1) Open your Web browser and go to drive.google.com.
- 2) Log in using your Google account.
- 3) On your Drive, click *New* → *Folder*.
- 4) You can name your folder whatever you like. For the purpose of this demonstration, let's call it *learn-pytorch*.
- 5) Go to the folder you just created and then click *New* → *More* → *Google Colaboratory* as shown in Figure 1.
- 6) This will take you to your Google Colab notebook. On the top left, an automatically generated name of the file will be displayed, which could be something like *Untitled0.ipynb*. The extension *.ipynb* stands for the Ipython notebook. Click on the name and rename it *linear*

regression using pytorch.ipynb.

- 7) On the left side of the top panel, below the main menu, you will see two buttons, '+ code' and '+ text', which you can use to create code or text snippets based on your requirements.
- 8) On the right side of the panel, you will be able to find the options to view the RAM and disk space allotted to you, as well as how much of it you have already used.
- 9) In order to test that everything is working properly, simply write the following code in the automatically generated first code cell:

```
print('Hello World')
```

- 10) Now press the *Shift* + *Return* key to execute the code or click the little *Play* sign at the beginning of the cell to execute the command. Your expected output should be displayed below the code as 'Hello World'.

And that's it; we are done with the basic setup! Let us start executing our code.

Deploying the model

The first thing is to check if PyTorch is already installed and if not, we need to install it. We will do this by running the following piece of code:

```
!pip3installtorch
```

Next, let us import the following libraries for the code execution:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import torch
from torch.autograd import Variable
```

In order to simplify things for the purpose of this demonstration, let us create some dummy data of the land's dimensions and its corresponding price with 20 entries. If you would like to directly read this data in the form of a

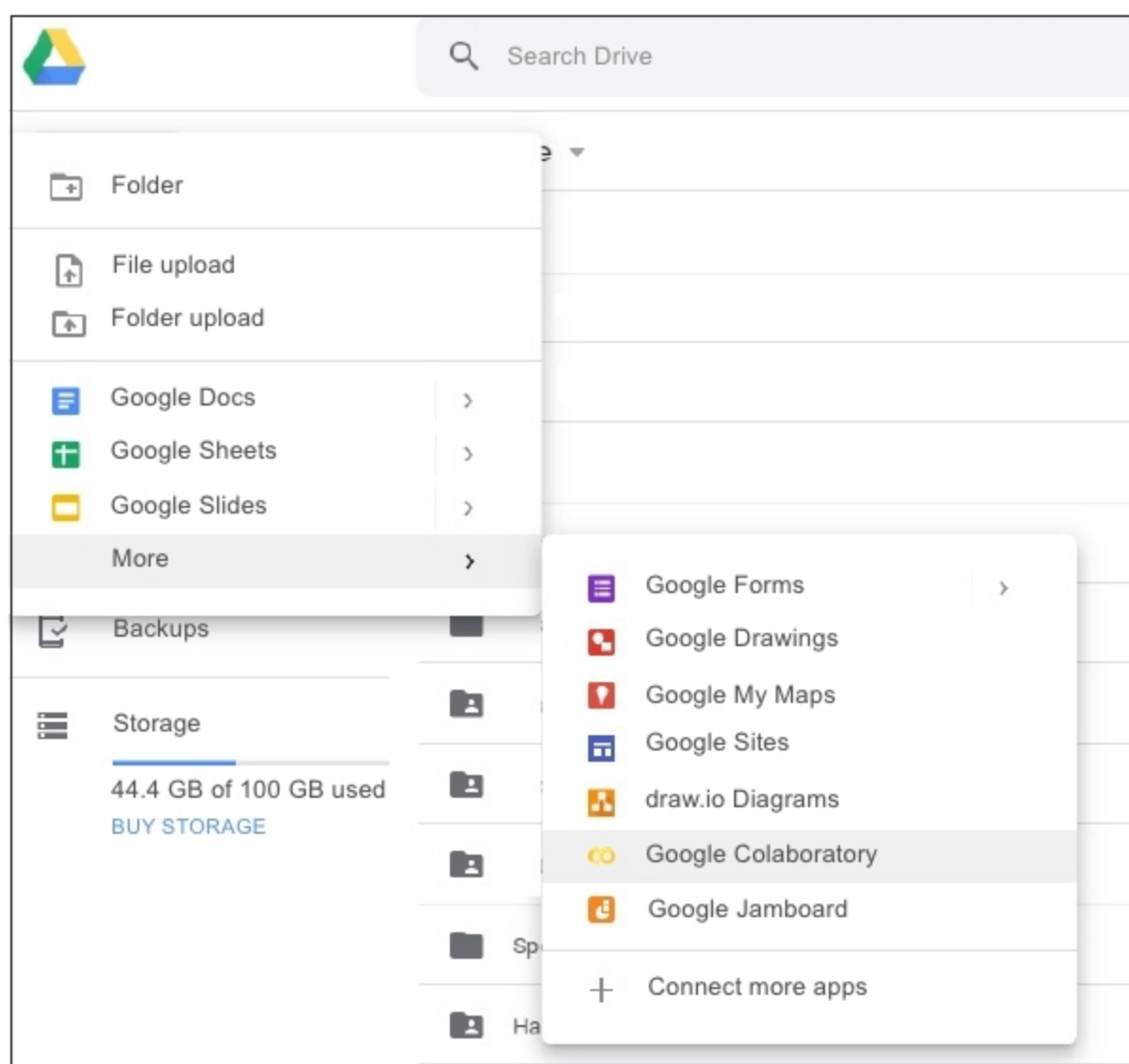


Figure 1: Creating a new Google Colab notebook in Google Drive